

Language	(LM for $P(S W)$, LM for $P(W)$)										ASR	Speaker Diarization			
Model (LM)	-		All n-gram		(LLM, n-gram)		(n-gram, LLM)		All LLM						
Matching	TS-match		BSD		BSD		BSD		BSD						
Metric	ΔSA	Δcp	ΔSA	Δcp	ΔSA	Δcp	ΔSA	Δcp	ΔSA	Δcp	WER	Miss	FA	CER	DER
CH-others	8.45	8.28	8.07	7.86	8.36	8.18	8.61	8.41	7.76	7.58	25.15	5.97	9.79	6.04	21.80
CH-109	5.05	4.97	3.70	3.64	3.70	3.63	4.42	4.30	3.24	3.16	23.07	5.55	4.39	1.44	11.38
AMI-MH-dev	8.22	8.20	3.47	3.45	3.49	3.47	3.52	3.50	3.45	3.42	23.68	15.89	4.02	1.02	15.89
AMI-MH-test	6.38	6.41	3.92	3.88	3.74	3.72	3.98	3.94	3.84	3.80	23.25	12.96	4.98	1.36	19.03

Table 1: Performance evaluation of multi-speaker ASR based on time-stamp (TS) matching and beam search decoding (BSD).

4. EXPERIMENTAL RESULTS

4.1. Experiment Settings

4.1.1. Pre-trained Models

- ASR:** We employ a Conformer-CTC model [26] that is implemented using the NeMo Toolkit [24]. The ASR model has approximately 122M parameters and 1024 tokens.
- Speaker Diarization:** We employ an improved version [20] of the MSDD model [19], which has 32M parameters. It builds upon Titanet [27], Transformer Encoder [28], and Clustering [29].
- LLM¹:** We use an LLM based on [23] which is a scaled version of GPT [31] model of 2B parameter size. Trained on a 1.1T token subset of the dataset curated by NVIDIA Data Collector (NDC) [30], containing 70% of English dataset, 15% of Stack v1.1 dataset [32] and 15% of multi-lingual datasets from Common Crawl [33].

4.1.2. Evaluation Metric

- WER:** The WER is determined using a hypothesis script generated from a single-channel mixed audio clip and a transcript that contains words in onset order. Note that this WER should be differentiated from the channel-specific or speaker-specific WER in other studies. We use text normalization tool from [34] for evaluation.
- SA-WER:** The Speaker-Attributed WER (SA-WER) [3] is a WER metric grounded in speaker mapping, as established by speaker diarization. SA-WER measures the WER by comparing hypothesis and reference transcripts for each specific speaker mapping.
- cpWER:** The concatenated minimum-permutation word error rate (cpWER) [35] is a metric designed to capture both ASR and diarization accuracy. cpWER is calculated by taking the minimum WER from concatenated transcripts of multiple speakers across all potential permutations.
- **Δcp and ΔSA :** delta-cpWER and delta-SA-WER follows the following relationship.

$$\Delta cp = cpWER - WER$$

$$\Delta SA = SA-WER - WER$$

Our evaluation is based on the assumption that Δcp and ΔSA reflect the diarization error in cpWER and SA-WER, respectively.

4.2. Datasets and Systems Overview

Table 1 shows the performance for each setup and dataset. For AMI-MH (Mixed Headset) [36] dataset, we use only-word version [37] of annotation. The Call Home American English Speech (CHAES, LDC97S42) is a corpus composed solely of English telephonic speech data. CH-109 is a subset that includes two speakers per session, whereas CH-others signifies the other sessions found within CHAES. We optimize α , β in Eq. (13) and context window length C in Eq. (10), beam-width for beam search decoding on CH-others and AMI-MH-dev then use the parameters for CH-109 and AMI-MH-test, respectively. For this parameter optimization, we employ Optuna [38]. The Diarization Error Rate (DER) is computed using a collar of 0.25 seconds while including overlaps.

¹Details regarding model training and datasets will be provided in [30]

In terms of systems, *TS-match* is a system that relies on time-stamps (TS) to match speaker diarization time-stamps with the decoded word time-stamps from ASR system. *All n-gram* and *All LLM* are systems where speaker probability and word probability values are calculated from the n-gram LM and the LLM, respectively. Conversely, *(LLM, n-gram)* and *(n-gram, LLM)* employ one model for speaker probability and another for word probability, in accordance with the provided notation. All experiments involving the LLM were conducted using the NVIDIA TESLA V100 GPU.

4.3. Analysis of Experimental Results

In Table 1, the performance of TS-matching is compared with four combinatory setups that integrate both n-gram and LLM. Regarding Δcp and ΔSA , it is crucial to highlight that speaker confusion resulting from speaker diarization results in a double error count because it manifests as an insertion for one speaker and a deletion for another. The proposed method improves the baseline system’s delta-SA-WER by up to 39.8%. We can deduce couple of findings: Firstly, the trend shows that applying same type of language model leads to a lower error rate. This is likely because the $P(W)$ and $P(S|W)$ are less prone to discrepancies. Secondly, LLM seems more effective in estimating $P(S|W)$, given that the average performance of applying LLM for $P(S|W)$ demonstrates superior performance over n-gram. We speculate that the performance difference stems from the fact that the n-gram model considers only a single speaker, while LLM processes the entire transcription, taking into account all speakers within the context window. This contextual understanding is a distinct advantage LLM holds over n-gram LM, providing a more nuanced estimation of the speaker. However, using LLM demands roughly 15 times more computational time during inference when compared to the n-gram LM. This discrepancy underscores the potential need for a trade-off, suggesting a hybrid (*LLM, n-gram*) configuration.

5. CONCLUSIONS

In this paper, we introduce a beam search decoding-based approach for applying a language model to speaker diarization. The proposed method offers a key advantage: it uses individually optimized models for ASR, diarization, and LLM. By training each model independently, we can leverage large-scale data sources specific to each domain. Moreover, even without fine-tuning the ASR, diarization, and LLM models, the proposed method achieves significant improvement over conventional approaches. This methodology can be seamlessly adapted to accommodate multilingual contexts by integrating multilingual ASR and LLM models—areas that have seen significant advancements recently. Looking forward, our future research will focus on several areas: Firstly, we intend to integrate beam search decoding for ASR and diarization by applying a single LLM which can achieve a more efficient system with enhanced accuracy. Secondly, we plan to integrate the ASR and diarization decoders to obtain more accurate timestamps alongside speaker logits, aiming for a streamlined multi-speaker ASR. Finally, we will explore ways to improve the model by introducing more sophisticated context, either by fine-tuning or prompt-tuning the LLM using domain-specific data.

6. REFERENCES

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